

WASP-36b

R-1386

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_220408 · created 2026-08-01T02:26:10+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459677.6865 +/- 0.0046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.13 +/- 0.048
Transit depth (Rp/Rs)^2	1.68 +/- 1.24 [%]
Orbital Inclination (inc)	87.83 +/- 2.9
Airmass coefficient 1 (a1)	0.127 +/- 0.059
Airmass coefficient 2 (a2)	1.21 +/- 0.48
Scatter in the residuals of the lightcurve fit is	44.11 %
Best Comparison Star	#1 - [605, 420]
Optimal Aperture	2.17
Optimal Annulus	3.0
Transit Duration (day)	0.0885 +/- 0.0096

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.13 $\pm$ 0.048 vs 0.1368 (deviation 0.13 σ)
Duration fit vs published	0.0885 d vs 0.0758 d (deviation 1.17 σ)
Mid-time O-C	11.1 min
Depth SNR	2.4
Residual scatter	44.11 %

Light curve

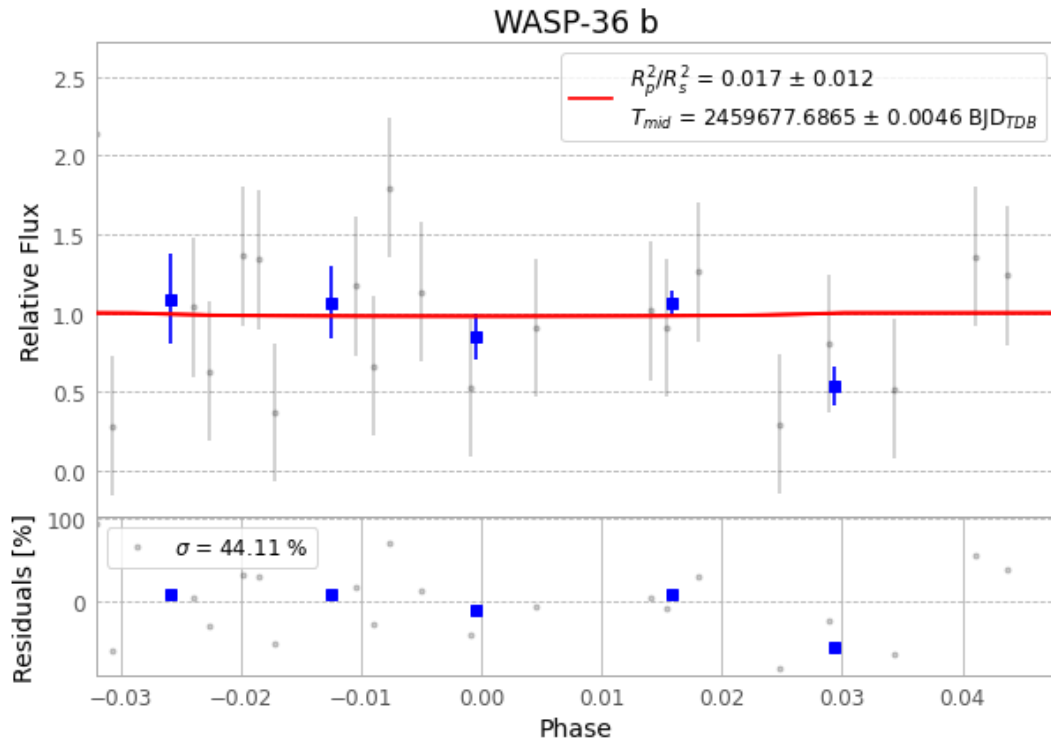


Figure 1 — FinalLightCurve_WASP-36 b_2022-04-07.png (SHA-256 7862890144e23149...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [402, 397], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 64ce135e79ad190e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), WASP-36220408030013.FITS → WASP-36220408060913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-220408.log (ac1ee74b53cf...), FinalLightCurve_WASP-36 b_2022-04-07.png (7862890144e2...), FinalLightCurve_WASP-36 b_2022-04-07.csv (d099572e0c03...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 02:26Z.